Module-2
SUSTAINABLE BUILT ENVIRONMENT

Sustainable Built Environment: Emerging materials: Types and Uses of Autoclaved Aerated Concrete (AAC) blocks, Bamboo, Recycled plastics, Material selection criteria, Durability, Sustainability, Smart City concept.

Green Building: Green building materials and rating systems IGBC, LEED, GRIHA (Green Rating for Integrated Habitat Assessment) for new buildings – Purpose - Key highlights - Point System with Differential weightage.

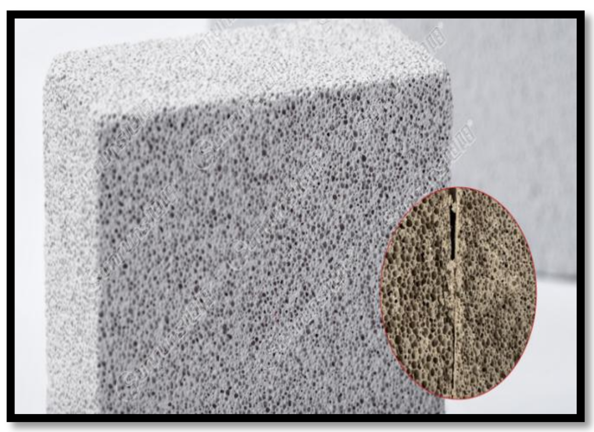
EMERGING MATERIALS IN SUSTAINABLE BUILT ENVIRONMENT

Emerging materials in civil engineering are new, innovative, or advanced materials that are developed to improve the strength, durability, sustainability, and performance of construction structures compared to traditional materials like concrete, steel, and bricks.

1. AUTOCLAVED AERATED CONCRETE (AAC) BLOCKS

Autoclaved Aerated Concrete (AAC) is lightweight precast foam concrete. These blocks are porous, reusable, non-toxic, renewable, and recyclable also. This superb building material has similar properties to wood. These green building materials provide an edge over wood for building structures without causing any decay, termite damage and combustibility.

Manufacturers combine cement, fine aggregates, water and an expansion agent in the manufacturing of these blocks. The expanding agent is primarily aluminum powder or paste, which reacts with other ingredients to create small air pockets. Manufacturers then pour the mixture into moulds and cure it in an autoclave, AAC blocks benefit from high pressure and steam to harden, a process known as steam curing. The final material obtained is a versatile and eco-friendly building material that is ideal for wider applications. AAC blocks are five times lighter than traditional concrete and contain approximately 80 percent air.



TYPES OF AAC BLOCKS

AAC blocks come in various types, each designed to fulfil unique construction requirements.

1. **Hollow AAC Blocks:** Designed for enhanced insulation and reduced weight, these blocks are perfect for low-load applications.
   - **Applications**: Best suited for partitions, boundary walls, and low-load applications.
   - **Features**: Enhanced thermal insulation and reduced material usage.
2. **Lightweight AAC Blocks:** Lightweight yet durable, these blocks are ideal for minimizing structural load in high-rise buildings.
   - **Applications**: Perfect for high-rise buildings where reducing structural load is crucial.
   - **Features**: Eases transportation and minimizes construction load.
3. **Fire Resistant AAC Blocks:** Specifically designed to provide maximum fire protection for sensitive structures.
   - **Applications**: Commonly used in buildings with high fire safety requirements.
   - **Features**: Enhanced safety and peace of mind.
4. **200 mm AAC Block:** Thick and robust, these blocks add stability and insulation to your construction projects.
   - **Applications**: Used in exterior walls and heavy-duty partitions.
   - **Features**: Provides robust thermal and sound insulation.
5. **100 mm AAC Block:** Thin, efficient, and lightweight, ideal for creating internal partitions.
   - **Applications**: Suitable for internal walls and non-load-bearing structures.
   - **Features**: Easy to handle and install, saving construction time.
6. **Long-lasting AAC Block:** Designed for durability, these blocks ensure minimal maintenance and extended lifespan.
   - **Applications**: Used in projects requiring high longevity and minimal maintenance.
   - **Features**: High resistance to environmental wear and tear.
7. **Reinforced AAC Blocks**
8. **Long Lasting AAC Blocks**
9. **Fly Ash AAC Blocks**

USES:

- **Load-Bearing Walls:** Ideal for constructing strong and durable load-bearing walls.
- **Non-Load-Bearing Walls:** Suitable for partition walls and interior structures.
- **Residential Projects:** Perfect for building homes due to their insulation and soundproofing properties.
- **Commercial Projects:** Used in office buildings and commercial spaces for efficient construction.
- **Industrial Projects:** Suitable for factories and warehouses due to their strength and fire resistance.
- **Exterior Walls:** Provide excellent thermal insulation, making them ideal for exterior walls.

- **High-Rise Buildings:** Lightweight nature makes them suitable for high-rise construction, reducing overall load.
- **Partition Walls:** Perfect for creating internal partitions with ease.
- **Fire-Resistant Structures:** Enhance safety with superior fire resistance.
- **Eco-Friendly Construction:** AAC blocks are a sustainable choice, contributing to green building practices.
- **Reduced Construction Time:** Precise dimensions and ease of installation speed up construction processes.

ADVANTAGES

1. **Lightweight Nature:** AAC blocks are up to 50% lighter than traditional bricks, reducing the overall dead load on a building. This feature makes them ideal for high-rise structures and retrofitting projects.
2. **Thermal Insulation:** The unique cellular structure of AAC blocks provides superior thermal insulation. This reduces heating and cooling costs, ensuring energy-efficient buildings.
3. **Eco-Friendly Material:** Made from natural resources like fly ash, lime, and cement, AAC blocks are a sustainable alternative to traditional construction materials. Their production process emits fewer greenhouse gases, contributing to a greener environment.
4. **Quick and Easy Installation:** AAC blocks are larger in size and lighter, making them quicker to install than conventional bricks. This speeds up the construction process and reduces labour costs.
5. **Fire and Pest Resistance:** AAC blocks are non-combustible and can withstand high temperatures, providing excellent fire protection. They are also pest-resistant, as their inorganic composition does not attract termites or rodents.
6. **Sound Insulation:** The porous structure of AAC blocks helps in absorbing sound, making them ideal for residential and commercial buildings requiring noise control.
7. **Durability and Strength:** Despite being lightweight, AAC blocks offer excellent compressive strength and durability. They can withstand harsh weather conditions and are resistant to cracking.

SUSTAINABILITY ASPECTS:

- Consumes fewer raw materials than conventional concrete.
- Produces less construction waste and lowers embodied energy.
- Recyclable and reusable for fill or sub-base material.

2. BAMBOO

Bamboo is an emerging sustainable material gaining traction in sectors like construction, textiles, and manufacturing due to its rapid growth, high strength, and renewable nature.

Why Bamboo is an Emerging Material

- **Sustainability & Renewability:** Bamboo is a fast-growing, renewable resource that can reach maturity in just three to five years, making it a sustainable alternative to slower-growing wood.
- **High Strength-to-Weight Ratio:** Bamboo possesses high tensile strength, making it suitable for load-bearing applications and a potential alternative to steel reinforcement in concrete.
- **Versatility:** It can be processed into a wide range of products, from structural components like panels and beams to everyday items like textiles, paper, and even food-related products.
- **Environmental Benefits:** Beyond its rapid growth, bamboo can also capture significant amounts of carbon dioxide, contributing to a smaller environmental footprint compared to conventional materials.

Application of bamboo in construction

1. **Bamboo for Foundations:** There is very limited use of bamboo as foundation material because when in contact with moisture laden surface they decay fast. However, this issue can be tackled to quite an extent though proper treatment using appropriate chemicals.
   The various types of foundations constructed with bamboo are:
   a) Bamboo which is in direct contact with ground surface.
   b) Bamboo fixed to rock or preformed concrete footings
   c) Composite bamboo or concrete columns
   d) Bamboo piles
2. **Walls Construction with Bamboo as a Building Material:** Bamboo is extensively used for construction of walls and partitions. Posts and beams are the main elements normally constructed with bamboo provide structural framework for walls. They positioned in a way to be able to withstand forces of nature. An infill is used between framing elements to add strength and stability to the walls.
3. **Scaffolding with Bamboo as a Building Material:** Due to advantageous properties of bearing heavy load bamboos are considered as one of the highly-endorsed materials for scaffolding even for tall structures. For the construction of scaffolding, bamboo extensions are obtained by lashing bamboo ends using several ropes.
4. **Reinforced Bamboo:** One of the most recent developments in bamboo construction is the use of reinforced bamboo, where bamboo is combined with other materials such as steel, concrete, or polymers to enhance its strength and durability. Reinforcement is crucial for using bamboo in larger, multi-story buildings and in regions with heavy loads or seismic activity.
5. **Bamboo Grids and Trusses:** Bamboo’s flexibility makes it a perfect material for structural grids and trusses that form the skeleton of a building. The strength-to-weight ratio of bamboo allows it to be used in long spans, providing open, airy interiors with minimal material usage. Bamboo grids are used in both roofing systems and facade designs

Advantages of Bamboo as a Building Material

- Tensile strength: Bamboo has higher tensile strength than steel because its fibers run axially.
- Fire Resistance: Capability of bamboo to resist fire is very high and it can withstand temperature up to 400^0C. This is due to the presence of high value of silicate acid and water.
- Elasticity: Bamboo is widely preferred in earthquake prone regions due to its elastic features.
- Weight of bamboo: Bamboos due to their low weight are easily displaced or installed making it very easier for transportation and construction.
- Unlike other building materials like cement and asbestos, bamboo poses no danger to health.
- They are cost effective and easy to use.
- They are especially in great demand in earthquake prone areas.

Disadvantages of Bamboo

Bamboos come with their own set of drawbacks such as:

- They require preservation
- Shrinkage: Bamboo shrinks much greater than any other type of timber especially when it loses water.
- Durability: Bamboo should be sufficiently treated against insect or fungus attack before being utilized for building purposes.
- Jointing: Despite prevalence of various techniques of jointing, structural reliability of bamboo is questionable**.**

Sustainability Aspect:

- Carbon-negative: Absorbs more CO_2 than emitted during processing.
- Renewable & biodegradable.
- Can be locally sourced, reducing transport emissions.

3. RECYCLED PLASTICS

Recycled plastics are an emerging class of sustainable construction materials that help reduce plastic waste, conserve natural resources, and lower the carbon footprint of buildings and infrastructure. These materials support circular economy principles by converting discarded plastics into useful construction products.

Recycled plastics used in construction are mainly obtained from **post-consumer** and **post-industrial plastic waste**. The waste plastic undergoes **collection, sorting, cleaning, and reprocessing** to produce construction-grade materials. The final products retain the versatility of conventional plastics while offering improved environmental performance. Commonly recycled plastics used in construction include:

- **Polyethylene (PE)**
- **Polypropylene (PP)**
- **Polyethylene Terephthalate (PET)**

Each type has specific properties that make it suitable for different construction applications such as road works, insulation, and building components.

Applications of Recycled Plastic in Construction

1. Base and Subbase for Road Construction: Recycled plastic waste (PW) is used as a partial replacement for natural aggregates in pavement base and subbase layers.

- Improves **shear strength, stiffness, and bearing capacity** of pavements
- Enhances pavement performance when used as plastic strips or granules
- Can be blended with demolition waste for road construction
- Slight reduction in stiffness and bearing capacity may occur due to the **smooth surface of plastic**, but overall performance remains acceptable

2. Recycled Plastic as Filler Material: Using plastic waste as filler is one of the simplest and most effective applications**.**

- Does not depend on the chemical properties of plastic
- Similar application as fillers in **cementitious composites and asphalt mixtures**
- Helps reduce material cost and plastic waste disposal

3. Wood Replacement (Plastic Lumber): Recycled plastic can be processed into **plastic lumber**, which behaves similarly to wood.

- Made from mixed plastic waste
- More durable than wood with high resistance to **weather, insects, and saltwater**
- Used for **railway sleepers, fences, benches, docks, and outdoor structures**
- High recycling cost and bulky nature limit widespread use

4. Door Panels: Plastic waste can be combined with wood or cellulose fibers to form eco-friendly door panels**.**

- Plastic pellets or powder mixed with wood flour
- Produces a **thermoformable wood–plastic composite**
- Suitable for interior door panels and partitions

5. Insulation Materials: Recycled plastic is used as an alternative to conventional insulation materials**.**

- Expanded Polystyrene (EPS) made from recycled plastic provides **thermal insulation**
- Reduces energy consumption in buildings
- Limitations include **low density, fire safety concerns, and transport difficulties**

6. Component in Cementitious Composites: Recycled plastic can be mechanically processed and used in concrete as aggregates or fillers.

- Reduces dependence on natural aggregate mining
- Lightweight nature helps in producing **lightweight concrete**
- Suitable mainly for **non-structural applications**
- Excess plastic content may reduce concrete strength, so usage must be controlled

7. Walls and Bricks: Recycled plastic can be used to manufacture wall panels and bricks.

- Plastic blocks are made by heating and moulding recycled plastic
- Suitable for **partition walls and non-load-bearing walls**
- Plastic bottles can be arranged like bricks for low-cost housing
- Sand–plastic blocks can be produced without water and show good durability
- Structural use is limited due to **low strength**

Sustainability Aspect:

- Reduces landfill and ocean pollution.
- Substitutes for virgin raw materials.
- Supports circular economy principles.

MATERIAL SELECTION CRITERIA

Selecting the right material is **crucial** for achieving sustainability, durability, cost-effectiveness, and functionality in modern buildings. For **emerging materials**, additional considerations such as lifecycle performance, recyclability, and local adaptability become even more important.

The process of selecting building materials goes beyond cost and aesthetic appeal. In modern construction, especially for smart and sustainable cities, the criteria are holistic.

1. **Durability** – Ability to resist wear, decay, and environmental effects over time.
2. **Strength / Structural Performance** – Ability to carry loads without failure.
3. **Environmental Impact / Sustainability** – Recycled content, embodied energy, low VOCs, life cycle impact.

4. **Cost & Availability** – Economic feasibility and local availability.
5. **Maintenance Requirement** – Ease of repair, cleaning, and upkeep over time.
6. **Aesthetic / Architectural Value** – Texture, color, finish, and visual appeal.
7. **Compatibility with Other Materials** – Chemical and physical compatibility in construction.
8. **Thermal / Acoustic Performance** – Insulation properties, fire resistance, sound absorption.
9. **Health & Safety** – Non-toxic, low-emission, fire-resistant materials.

DURABILITY CRITERIA

Durability is a material's ability to maintain its integrity and function over time and is assessed by considering:

1. **Environmental Conditions**: Exposure to moisture, rain, humidity, freeze–thaw cycles, UV radiation, wind, and temperature fluctuations.
2. **Mechanical Stresses**: Load-bearing capacity, abrasion, impact, vibration, or repeated use.
3. **Chemical Resistance**: Resistance to corrosion, acids, alkalis, salts, or pollutants (e.g., reinforced concrete exposed to chlorides).
4. **Biological Resistance**: Resistance to termites, fungi, mold, algae, or insects.
5. **Maintenance Needs**: Materials that require less frequent repair or replacement are considered more durable.

SUSTAINABILITY CRITERIA

Sustainable material selection focuses on reducing the environmental impact throughout the entire life cycle:

1. **Embodied energy:** The total energy consumed in the extraction, manufacturing, and transportation of a material.
2. **Recycled content and recyclability:** The percentage of recycled material in a product and its potential for future reuse or recycling.
3. **Local sourcing:** Sourcing materials locally reduces transportation costs and carbon footprint.
4. **End-of-life considerations:** How easily a material can be disassembled, reused, or recycled at the end of a building's life.
5. **Renewability:** Choosing materials from renewable resources that can be replenished quickly.
6. **Low toxicity:** Selecting materials that do not emit harmful volatile organic compounds (VOCs) or other toxins, which improves indoor air quality.

SMART CITY CONCEPT

A **Smart City** is an urban system that integrates **Information and Communication Technology (ICT)**, **Internet of Things (IoT)**, and **data analytics** to manage city resources efficiently, enhance service delivery, and improve the quality of life of citizens. A **Smart City** uses **digital technology, data, and sustainable infrastructure** to improve the quality of life for its residents while optimizing resource efficiency and minimizing environmental impact.

Unlike traditional cities, smart cities rely on **real-time data**, **automation**, and **intelligent decision-making systems** to address challenges such as population growth, traffic congestion, pollution, water scarcity, and energy demand.

Core Objectives of a Smart City:

1. **Sustainability** – Reduce carbon footprint, energy consumption, and waste.
2. **Efficiency** – Optimize transportation, utilities, and building operations.
3. **Safety and Security** – Advanced surveillance, disaster management, and resilient infrastructure.
4. **Connectivity and Innovation** – Internet of Things (IoT), sensors, and data-driven decision-making.
5. **Citizen-Centric Services** – Smart healthcare, education, mobility, and governance.

Key Components:

1. **Smart Governance** –focuses on transparency, efficiency, and citizen engagement.
   - E-governance platforms for service delivery
   - Online tax payment and grievance redressal
   - Open data portals
   - Real-time monitoring of civic services
   - Participatory decision-making through mobile apps

2. **Smart Mobility** – Electric vehicles, traffic management, shared mobility , Improved public transportation, intelligent traffic management, and pedestrian-friendly pathways.Transportation systems are optimized to reduce congestion and emissions.
   - Intelligent Traffic Management Systems (ITMS)
   - Adaptive traffic signals
   - GPS-enabled public transport
   - Smart parking systems
   - Promotion of electric vehicles (EVs) and non-motorized transport

3. **Smart Environment** – Environmental sustainability is a key component of smart cities.
   - Air and noise pollution monitoring
   - Green buildings and energy-efficient designs
   - Urban green spaces and parks
   - Climate-resilient infrastructure

- Disaster prediction and early warning systems

4. **Smart Water Management** – Smart water systems ensure sustainable use and conservation of water resources.
   - Smart water meters
   - Leakage detection and pressure monitoring
   - Rainwater harvesting systems
   - Wastewater treatment and reuse
   - Real-time water quality monitoring

5. **Smart Energy**– Smart energy systems aim at efficient production and consumption.
   - Smart grids for real-time energy monitoring
   - Smart meters for consumers
   - Integration of renewable energy sources
   - Energy-efficient public lighting
   - Demand-side energy management

6. **Smart Buildings/ infrastructure**: Smart infrastructure uses sensors, automation, and monitoring systems to improve reliability and safety.
   - Smart roads with embedded sensors
   - Intelligent street lighting (automatic ON/OFF)
   - Structural health monitoring of bridges and buildings
   - Smart drainage and flood control systems

7. **Smart Living** – Improved healthcare, education, and affordable housing.
8. **Smart People** – Digital literacy, skill development, and active citizen participation.

Challenges in Smart City Development

- High cost of implementation
- Integration of legacy infrastructure with modern technology
- Public awareness and participation
- Cyber security threats
- Coordination between government agencies

Technologies Involved in Smart City Development

Smart city development relies on advanced technologies to improve urban services, sustainability, and quality of life. The key technologies involved are:

1. Internet of Things (IoT): IoT connects sensors and devices to collect real-time data from city infrastructure such as traffic signals, water pipelines, waste bins, and street lights. This enables efficient monitoring and control of city services.

2. Information and Communication Technology (ICT): ICT forms the backbone of smart cities by enabling digital communication, e-governance, online public services, and real-time information sharing between authorities and citizens.

3. Artificial Intelligence (AI) and Machine Learning: AI is used for intelligent decision-making such as traffic prediction, energy demand forecasting, and predictive maintenance of infrastructure.

4. Big Data Analytics: Big data technology analyzes large volumes of data generated by sensors and citizens to support urban planning, resource management, and policy decisions.

5. Cloud Computing: Cloud platforms provide scalable data storage, processing, and access to city data, reducing infrastructure costs and improving service efficiency.

6. Geographic Information System (GIS) and GPS: GIS and GPS are used for mapping, land-use planning, navigation, infrastructure management, and disaster response.

7. Smart Sensors and Automation: Sensors monitor air quality, water quality, noise levels, and structural health, while automation systems enable efficient operation of lighting, traffic, and utilities.

Green Building: Green building materials and rating systems IGBC, LEED, GRIHA for new buildings – Purpose - Key highlights - Point System with Differential weightage.

GREEN BUILDING

Green building is a resource-efficient construction and development approach that considers environmental impact and human health. In green building projects, sustainability is incorporated throughout the building's lifecycle, from planning to demolition.

Green building concepts include environmentally conscious site selection, practices to facilitate and improve energy efficiency, water efficiency and indoor environmental quality, and efforts to limit carbon emissions. Successful green building projects often meet widely adopted green building standards, such as the Leadership in Energy and Environmental Design (LEED) framework.

Importance of Green Buildings

Some 38% of global energy-related carbon emissions stem from building construction and building operations.[3] With greenhouse gas emissions known to be the leading cause of anthropogenic global warming, sustainable design and sustainable building are emerging as critical efforts to mitigate the impacts of climate change. In addition to environmental impact, green building programs also yield other benefits such as:

- Lowering building operating costs and utility bills through reduced energy consumption and water use.
- Enhancing public health, quality of life and productivity through more comfortable and healthier indoor environments.

- Helping developers and building owners meet environmental, social and governance (ESG) standards.
- Increasing the likelihood of buildings becoming high-performing assets, with green buildings often achieving higher market value than comparable structures.

Key Features of Green Buildings

- Solar panels and renewable energy systems
- High thermal insulation
- Green roofs and vertical gardens
- Smart energy and water monitoring systems
- Waste segregation and recycling systems

Green Building Rating Systems

These systems evaluate energy efficiency, water use, materials, and environmental impact. Green buildings are assessed using rating systems such as:

- **IGBC** (Indian Green Building Council)
- **LEED** (Leadership in Energy and Environmental Design)
- **GRIHA** (Green Rating for Integrated Habitat Assessment – India)

GREEN MATERIALS

Green materials are environmentally friendly construction materials that minimize negative impacts on the environment and human health throughout their life cycle—from extraction to disposal. The main characteristics are discussed below.

Characteristics of Green Materials

1. Low Environmental Impact: Green materials cause minimal damage to the environment during extraction, manufacturing, use, and disposal. They generate less pollution and reduce greenhouse gas emissions.
2. Renewable or Recyclable: Green materials are either **renewable** (such as bamboo and timber) or **recyclable** (such as steel, aluminum, and plastics), reducing the depletion of natural resources.
3. Low Embodied Energy; These materials require less energy for production, transportation, and installation compared to conventional materials, resulting in lower carbon emissions.
4. Use of Recycled Content: Green materials often contain recycled or waste materials such as fly ash, slag, recycled aggregates, and recycled plastics, which help in effective waste management.
5. Durability and Long Life: They are durable and have a long service life, reducing the need for frequent replacement and maintenance, thereby conserving resources.

6. Non-Toxic and Low VOC Emissions: Green materials emit low or no volatile organic compounds (VOCs), improving indoor air quality and ensuring occupant health and safety.
7. Energy Efficiency: Green materials contribute to thermal insulation, reflectivity, or shading, reducing heating and cooling energy requirements in buildings.
8. Water Efficiency: Some green materials support water conservation through reduced water usage during production or through permeability and water-harvesting capabilities.
9. Local Availability: Locally sourced materials reduce transportation energy and associated emissions while supporting the local economy.
10. Reusability and Biodegradability: Green materials can be reused or naturally decomposed without causing environmental harm at the end of their life cycle.

IGBC (INDIAN GREEN BUILDING COUNCIL)

The green concepts and techniques in the building sector can help address national issues like water efficiency, energy efficiency, and reduction in fossil fuel use in commuting, handling of consumer waste and conserving natural resources. Most importantly, these concepts can enhance occupant health, happiness and well-being.

The Indian Green Building Council (IGBC) has launched IGBC- Green New Buildings rating system to address the National priorities. This rating programme is a tool which enables the designer to apply green concepts and reduce environmental impacts that are measurable. The rating programme covers methodologies to cover diverse climatic zones and changing lifestyles.

Scope

IGBC Green New Buildings rating system is designed primarily for new buildings. New Buildings include offices, IT parks, banks, shopping malls, hotels, airports, stadiums, convention centers, libraries, museums, etc., Building types such as residential, factory buildings, schools will be covered under other IGBC rating programmes. IGBC Green New Buildings rating system is broadly classified into two types:

- **Owner-occupied buildings** are those wherein 51% or more of the building's built-up area is occupied by the owner.
- **Tenant-occupied buildings** are those wherein 51% or more of the building's built-up area is occupied by the tenants.

Based on the scope of work, projects can choose any of the above options. The project team can evaluate all the possible points to apply under the rating system using a suitable checklist (Owner-occupied buildings and Tenant-occupied buildings). The project can apply for IGBC Green New Buildings rating system certification, if it can meet all mandatory requirements and achieve the minimum required points.

Purpose - Why IGBC Green New Buildings exists

- To **reduce the environmental footprint** of newly constructed buildings in India (energy, water, waste, materials and land).
- To **raise occupant comfort & health** (ventilation, daylight, low-emission materials).
- To **set measurable benchmarks** so projects can show verified savings and qualify for incentives.
- To align green building practice with **national priorities** (water stress, energy efficiency, waste management, reduced fossil fuel use).

Benefits of IGBC

- IGBC helps in **reducing energy and water consumption** in buildings by promoting efficient design and sustainable practices.
- It supports the use of **eco-friendly and recycled materials**, which conserves natural resources and minimizes environmental degradation.
- IGBC-certified buildings offer improved indoor air quality, thermal comfort, and overall **occupant health and productivity.**
- The rating system helps in **lowering operating and maintenance costs**, resulting in long-term economic benefits.
- IGBC certification **enhances the market value and credibility** of buildings and provides recognition from government and industry bodies.
- Overall, IGBC plays a significant role in **promoting sustainable development** and environmentally responsible construction practices in India.

Key highlights (what makes the IGBC New Buildings rating distinctive)

- **Two project tracks**: Owner-occupied and Tenant-occupied (projects pick the path matching occupancy).
- **Mandatory requirements + credits**: Several prerequisites must be met first (e.g., minimum energy efficiency, ventilation, waste segregation). Credits are then earned for voluntary measures. Failure to meet prerequisites disqualifies the project.
- **Modules cover full lifecycle topics**: Sustainable architecture & design, Site selection & planning, Water conservation, Energy efficiency, Building materials & resources, Indoor environmental quality (IEQ), Innovation & development.
- **Performance and prescriptive mix**: Some credits are performance-based (e.g., % energy savings) while others are prescriptive (e.g., provide rainwater harvesting).
- **Third-party review and site visit**: IGBC conducts document review and site audits before awarding the rating

Main Evaluation Categories

IGBC Green New Buildings rating system® addresses green features under the following categories:

1. Sustainable Architecture & Design
2. Site Selection & Planning
3. Water Conservation
4. Energy Efficiency
5. Building Materials & Resources
6. Indoor Environmental Quality
7. Innovation & Design Process

The guidelines detailed under each mandatory requirement & credits, enables the design and construction of new buildings of all sizes and types. Different levels of green building certification are awarded based on the total credits earned. However, every green new building should meet certain mandatory requirements, which are non-negotiable.

Point System with Differential Weightage

IGBC follows a **point-based rating system**, where credits are assigned **different weightages** based on their environmental impact. Typical Weightage Distribution (Indicative):

Category	Approximate Weightage
Energy Efficiency	25–30%
Water Conservation	15–20%
Materials & Resources	10–15%
Indoor Environmental Quality	10–15%
Sustainable Site Planning	10–15%
Innovation & Design	5–10%

IGBC Certification Levels: Based on total points scored, buildings are rated as:

Certification Level	Points Range (out of 100)
Certified	50–59
Silver	60–69
Gold	70–79
Platinum	80–100

IGBC Evaluation Process

1. The project is first **registered with IGBC** under the appropriate rating system based on building type and occupancy (owner-occupied or tenant-occupied).
2. The project team must **comply with mandatory prerequisites**, such as minimum energy efficiency, water conservation measures, ventilation, and waste management. Failure to meet these requirements disqualifies the project.
3. After meeting prerequisites, the project earns **credits by adopting additional green measures** in categories like sustainable site planning, water conservation, energy efficiency, materials and resources, indoor environmental quality, and innovation.
4. IGBC uses a **point-based system with differential weightage**, giving higher importance to energy efficiency and water conservation.
5. The project team submits **design and performance documents** through the IGBC portal for evaluation.
6. IGBC conducts a **third-party document review** to verify compliance with prerequisites and credits.
7. **Site visits and audits** are carried out to confirm on-ground implementation of green measures.
8. Based on the **total points scored**, the project is awarded an IGBC rating such as **Certified, Silver, Gold, or Platinum**.
9. The final certification is issued only after successful **verification and approval** by IGBC.

LEED (LEADERSHIP IN ENERGY AND ENVIRONMENTAL DESIGN)

A G**lobal certification** by the **U.S. Green Building Council (USGBC)**, adapted in India as **LEED India** by IGBC. It benchmarks **sustainable design, construction, and operation** performance. LEED is the most widely recognized green building rating system in the world. LEED certification offers a framework for healthy, efficient, and cost-effective green buildings, providing environmental and social benefits. It serves as a globally recognized symbol of sustainability leadership, supported by a committed community of organizations and individuals driving market transformation

Scope of LEED

- LEED is an internationally recognized green building rating system developed by the U.S. Green Building Council (USGBC).
- Its scope includes the planning, design, construction, operation, and maintenance of green buildings.
- LEED certification is applicable to residential, commercial, institutional, industrial, and mixed-use buildings, as well as existing buildings, interiors, and large developments.
- It evaluates buildings across their entire life cycle to ensure environmental sustainability and resource efficiency.

Purpose — Why LEED for new buildings

- Provide a **consistent, measurable framework** that reduces a building's environmental impact (energy, water, materials, land and indoor environment) over its lifecycle.
- Drive **market transformation** by rewarding high-performance design, verified operation and innovation.
- Deliver **occupant health & productivity** improvements (IEQ) while lowering operating costs and carbon emissions.

Benefits of LEED

1. LEED helps in significantly reducing energy and water consumption in buildings, thereby lowering operating costs.
2. It promotes the use of sustainable and recycled materials, reducing the depletion of natural resources.
3. LEED-certified buildings provide better indoor air quality and comfort, leading to improved occupant health and productivity.
4. The certification enhances the market value and global recognition of buildings.
5. LEED also supports environmental protection by reducing carbon emissions and promoting sustainable urban development.

Key highlights — what makes LEED BD+C (New Construction) distinctive

- **Prerequisites + credits model:** Projects must satisfy a set of prerequisites (mandatory baseline requirements) and then earn credits to accumulate points. **Multiple, harmonized categories:** Credits are grouped into categories that reflect where buildings make the biggest environmental difference (e.g., Energy, Water, Materials, Indoor Environmental Quality).
- **Performance + prescriptive approaches:** Many credits are performance-based (e.g., modeled energy savings) while others are prescriptive checklists. LEED v4.1 added more performance/prerequisite options.
- **Bonus points:** Innovation and Regional Priority credits provide up to **10 additional points** (6 + 4), so the maximum possible is **110 points**.
- **Third-party review & verification:** Documentation review and on-site verification ensure credibility and measured performance.

Point System and Certification Levels

Certification Level	Points Range (out of 110)
Certified	40–49
Silver	50–59
Gold	60–79
Platinum	80+

Differential Weightage

Category	Approximate Points	Weightage Importance
Energy and Atmosphere (EA)	~30–35 points	Highest
Location & Transportation (LT)	~15–20 points	High
Sustainable Sites (SS)	~10 points	Moderate
Water Efficiency (WE)	~10–12 points	Moderate
Materials and Resources (MR)	~10–13 points	Moderate
Indoor Environmental Quality (IEQ)	~15–16 points	High
Innovation (IN)	~5–6 points	Low
Regional Priority (RP)	~4 points	Special

LEED Rating System – Evaluation Process

1. **Project Registration**
   The project is first registered with the U.S. Green Building Council (USGBC) under the appropriate LEED rating system based on building type.
2. **Selection of Rating System**
   The project team selects the suitable LEED category such as New Construction, Existing Buildings, Homes, or Interior Design.
3. **Compliance with Mandatory Prerequisites**
   All mandatory prerequisites related to energy efficiency, water conservation, ventilation, and site sustainability must be fulfilled. Failure to meet any prerequisite disqualifies the project.
4. **Earning Optional Credits**
   After meeting prerequisites, the project earns points by achieving optional credits across LEED categories like sustainable sites, water efficiency, energy and atmosphere, materials and resources, indoor environmental quality, and innovation.
5. **Documentation and Submission**
   Required design and performance documents, drawings, calculations, and reports are uploaded through the LEED Online platform.
6. **Third-Party Review**
   Independent LEED reviewers assess the submitted documents to verify compliance and performance.
7. **Review Comments and Clarifications**
   The project team responds to reviewer comments and submits clarifications or revisions if required.
8. **Final Scoring and Certification**
   Total points are calculated, and the project is awarded a LEED certification level—Certified, Silver, Gold, or Platinum.

GRIHA (GREEN RATING FOR INTEGRATED HABITAT ASSESSMENT)

India's **national green rating system**, jointly developed by **TERI** and the **Ministry of New and Renewable Energy (MNRE)**. It measures a building's environmental performance over its entire life cycle. GRIHA aims to promote **sustainable, climate-responsive, and resource-efficient buildings** by reducing energy and water consumption, minimizing waste generation, and improving indoor environmental quality. The rating system assesses a building over its **entire life cycle**, from site planning and construction to operation and maintenance.

Scope of GRIHA

The scope of GRIHA **covers the promotion and assessment of** sustainable building practices across the entire life cycle of buildings in India**.**

- It is applicable to a wide range of building types including **residential, commercial, institutional, and government buildings**, as well as large developments such as **campuses and townships**.
- GRIHA evaluates buildings from the **pre-construction stage (site selection and planning)** through **construction** and **operation & maintenance**, ensuring holistic sustainability.
- The rating system is especially suited to **Indian climatic conditions and local construction practices**, making it relevant for both **new buildings and major renovations**.
- GRIHA also supports national sustainability objectives by encouraging **low-carbon development, resource conservation, and improved occupant comfort**, thereby contributing to environmentally responsible and energy-efficient built environments.

Purpose — why GRIHA for new buildings?

- To evaluate and improve the environmental performance of buildings across their lifecycle (site, design, construction and commissioning) and provide a national benchmark for "green" buildings in India.
- To drive resource efficiency (energy, water, materials), protect site ecology, ensure occupant health & comfort, and encourage social & economic benefits (worker safety, accessibility).

Key highlights — what makes GRIHA distinctive

- **India-tuned**: GRIHA adapts to Indian climatic, regulatory and water/energy contexts (e.g., strong emphasis on water, social measures).
- **Performance-** oriented 100-point system with a small Innovation/bonus pool (+5). The main rating adds to 100 points; Innovation can add up to 5 extra points.
- **Eligibility & thresholds:** Typical eligibility starts at built-up area > 2,500 m²; ratings are expressed as 1-star to 5-star based on points earned.
- **Three site visits & verification:** GRIHA requires due-diligence site visits (early, mid, final) and documentary evaluation before awarding rating.
- **Lifecycle concepts integrated:** LCA/LCCA (life-cycle analysis/costing), commissioning and

performance metering are explicit criteria.

Main Evaluation Criteria

1. **Site Planning**
2. **Conservation & Efficient Use of Resources**
   - Energy, Water, and Waste Management
3. **Building Operation & Maintenance**
4. **Indoor Environmental Quality**
5. **Innovation and Design**

Point System and Rating Levels

Certification Level	Points Range (out of 100)
1 Star	25–40
2 Star	41–55
3 Star	56–70
4 Star	71–85
5 Star	86+

Differential Weightage

Assessment Category	Focus Area	Approximate Weightage (%)
Sustainable Site Planning	Site selection, land use, microclimate	10–15
Construction Management	Soil conservation, waste control	5–10
Energy Efficiency	Building envelope, HVAC, lighting, renewables	25–30
Water Conservation	Rainwater harvesting, reuse, efficiency	15–20
Materials & Resources	Low-embodied energy, recycled materials	10–15
Solid Waste Management	Segregation, recycling, disposal	5–10
Indoor Environmental Quality (IEQ)	Thermal comfort, ventilation, daylight	10–15
Innovation & Design	Innovative green practices	5–10

GRIHA Evaluation Process

1. **Project Registration:**
   The project is first registered under GRIHA after selecting the appropriate rating variant based on building type and size.
2. **Compliance with Mandatory Criteria:**
   Certain criteria are mandatory, such as minimum energy efficiency, water conservation measures, and waste management practices. Non-compliance leads to disqualification.
3. **Assessment across Sustainability Criteria:**
   GRIHA evaluates buildings based on various sustainability aspects including site planning, energy efficiency, water management, materials, waste management, and indoor environmental quality.
4. **Credit-Based Point System:**
   Each criterion carries specific points. Points are awarded based on the extent of compliance and performance.
5. **Performance-Oriented Evaluation:**
   GRIHA emphasizes actual building performance rather than only prescriptive measures, particularly for energy and water efficiency.
6. **Submission of Documentation:**
   Project teams submit design details, calculations, drawings, and supporting documents through the GRIHA online portal.
7. **Third-Party Review:**
   Independent GRIHA evaluators review submitted documents to ensure transparency and accuracy.
8. **Site Inspection:**
   Physical site visits are conducted to verify implementation of proposed green measures.
9. **Final Scoring and Rating:**
   Based on total points achieved, the building is awarded a star rating.

COMPARATIVE SUMMARY OF RATING SYSTEMS

Aspect	IGBC	LEED	GRIHA
Full Form	Indian Green Building Council	Leadership in Energy and Environmental Design	Green Rating for Integrated Habitat Assessment
Developed By	CII – Indian Green Building Council	U.S. Green Building Council (USGBC)	TERI & Ministry of New and Renewable Energy (MNRE), Govt. of India
Year of Launch	2001	1998	2007
Applicability	India-focused	Global	India-focused
Climatic	Indian climatic	Global standards	Indian climatic zones

Aspect	IGBC	LEED	GRIHA
Consideration	conditions		
Nature of System	Voluntary	Voluntary	Mandatory for some govt. projects
Building Types	New, existing, homes, factories, campuses	New, existing, interiors, neighborhoods	Buildings, habitats, townships
Evaluation Approach	Prerequisites + credits	Prerequisites + credits	Mandatory criteria + points
Focus Areas	Energy, water, materials, IEQ	Energy, water, materials, site, IEQ	Energy, water, site, waste, comfort
Performance Emphasis	Moderate	High	Very high
Third-Party Review	Yes	Yes	Yes
Site Verification	Yes	Yes	Yes
Rating Levels	Certified, Silver, Gold, Platinum	Certified, Silver, Gold, Platinum	1 to 5 Star
Government Recognition (India)	High	Moderate	Very High
Ease of Adoption	Easy	Moderate	Relatively complex

Case Study: Infosys Campus, Pocharam, Hyderabad, India

1. Introduction

The **Infosys Pocharam Campus** in Hyderabad is a prime example of a **Smart and Green Building development** in India. It demonstrates how **information technology**, **energy efficiency**, and **sustainable design principles** can come together to create a **Smart City–aligned, environmentally responsible complex**.

2. Overview

Parameter	Details
Project Name	Infosys Pocharam SEZ Campus
Location	Pocharam, Hyderabad, Telangana, India
Developer	Infosys Limited

Parameter	Details
Architects	Hafeez Contractor & Infosys Design Team
Site Area	~130 acres
Green Building Rating	IGBC Platinum Certified (New Construction)
Certification Year	2012 (Phase 1)
Occupancy	Over 25,000 employees

3. Purpose

- To create a **self-sustained, smart, and eco-friendly IT campus**.
- To align with **Smart City principles**—efficient resource management, smart mobility, and sustainable infrastructure.
- To achieve **zero waste discharge**, **energy-efficient operations**, and **enhanced occupant comfort**.

4. Smart City Features

- **Smart Infrastructure** – Automated lighting, HVAC, and water management systems.
- **Smart Energy Management** – Real-time monitoring through Building Management Systems (BMS).
- **Smart Transport** – Electric vehicle charging stations and internal shuttle services.
- **Waste-to-Energy** systems and **rainwater harvesting**.
- **Integrated ICT systems** for maintenance and facility monitoring.

5. Green Building Highlights (IGBC Platinum Certified)

a) Sustainable Site Planning

- Over **70% of open area** landscaped with native and drought-tolerant species.
- **Minimal heat island effect** through reflective roofing and shaded pathways.
- Stormwater managed through **permeable paving** and **recharge pits**.

b) Energy Efficiency

- **Energy savings > 40%** compared to conventional buildings.
- Use of **high-performance glass**, **daylighting**, and **motion-sensor–based lighting controls**.
- **Solar photovoltaic panels** provide part of the building's electricity.

c) Water Efficiency

- **100% wastewater treatment** and reuse for landscaping and flushing.
- **Rainwater harvesting** structures capture more than **90% of rooftop runoff**.
- **Low-flow fixtures** reduce water consumption by 40%.

d) Materials and Resources

- Use of **fly ash–based concrete**, **recycled steel**, and **locally sourced materials**.
- More than **95% of construction waste** was reused or recycled on site.

e) Indoor Environmental Quality

- **CO_2 monitoring**, **low-VOC paints**, and **daylight optimization** ensure healthy indoor air.
- High acoustic and thermal comfort levels.

6. IGBC Point System (Achieved)

Category	Max Points	Points Achieved
Sustainable Sites	15	13
Water Efficiency	10	9
Energy Efficiency	25	23
Materials & Resources	10	8
Indoor Environmental Quality	15	13
Innovation & Design Process	10	9
Total	**85 / 100**	**Platinum Rating**

Differential Weightage: Energy & Water Efficiency categories contributed the highest points due to their major impact in India’s climatic context.